

Research Companion

A team of AI helpers that reads your research paper and checks its own answers before showing them to you.

This guide assumes you have never used the tool - or anything like it. Every term is explained the first time it appears. By the end you will know what the tool does, how to run it on your own paper in about ten minutes, how to read every result it gives you, and what to do when something goes wrong.

Free and open source (MIT) - github.com/Laraib-Hasan/Research-Companion

1. What is Research Companion?

When you finish writing a research paper, three worries keep you up at night: **Are all my references real and correct? Is my idea actually new, or did someone publish it already? And what will reviewers attack?**

Research Companion is a free program that answers those questions *before* you submit. You give it your paper (a PDF or an arXiv link). A team of specialized AI helpers - we call them **agents** - each examines one aspect: one checks every reference against real scholarly databases, one hunts for prior work similar to your claims, one judges how novel each claim is, one scores how confident you should be, and one even drafts replies to reviewer criticism.

The one thing that makes it different from asking a chatbot: **every answer comes with a receipt**. If the tool says a citation is fake, that is because a live database lookup found no such paper - not because an AI "felt" it was fake. If it quotes your paper, it re-checks that the quote really appears in your paper, word for word. Answers it cannot verify are flagged, never hidden.

2. The three ideas behind it

Idea 1 - Agents (specialists, not one generalist). Instead of one AI doing everything, the work is split among small specialists that each do one job well. They run at the same time where possible, and if one fails (for example, a database is temporarily busy), the others simply carry on - you get a partial result instead of a crash.

Idea 2 - The knowledge graph (the tool's memory). When the tool reads your paper, it does not just store the text. It extracts the *concepts*, *methods*, *datasets*, and *claims*, and connects them into a network - a knowledge graph. If two papers mention the same method, that method becomes one shared node connecting them. This is how the tool "remembers" and reasons across many papers at once.

Idea 3 - Verification (receipts for everything). Whenever an AI produces text, the tool independently checks it: quotes are matched against your paper character by character; references are looked up in CrossRef and OpenAlex (two large, free scholarly databases); confidence scores are computed by a plain formula you can read - no AI involved. Anything that fails a check gets a visible warning label.

3. Meet the helpers

Helper	What it does for you
Citation detective	Looks up every reference in your bibliography in live scholarly databases. Tells you which are confirmed real, which look wrong (bad DOI, wrong authors), and which cannot be found at all - the classic sign of an AI-invented citation.
Prior-art librarian	Searches the literature for papers related to yours, so the novelty judgment is grounded in what actually exists.
Novelty judge	Extracts what your paper claims as new, compares each claim against the retrieved related work, and gives a verdict: novel, incremental, overlaps, or anticipated - with the closest competing paper named.
Confidence scorer	Gives each claim a score like 0.72 +/- 0.15. No AI here - just a formula over three visible signals, so you can always see WHY the score is what it is.
Benchmark scout	Suggests datasets and benchmarks that related papers evaluate on - useful when a reviewer asks 'why didn't you compare on X?'

Rebuttal writer	You paste reviewer comments; it drafts polite point-by-point replies that quote only real passages from your paper. Any quote it cannot verify is flagged for you to check, never silently shipped.
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4. Your first ten minutes (follow along)

You need: Python 3.10+, a terminal, and (for the AI parts) an API key from OpenAI or Anthropic. There is also a zero-key demo at the end of this section if you want to try the tool without any account at all.

Step 0 - Install (once):

```
git clone https://github.com/Laraib-Hasan/Research-Companion
cd Research-Companion
pip install -e "[demo]"
```

Step 1 - Add your paper. This just stores it locally and gives it an ID (shown after 'local:'). Nothing is sent anywhere yet:

```
research-companion add my-draft.pdf --title "My Paper" --authors "Me" --year 2026
```

Step 2 - Ask what it will cost. Build the habit: this command tells you the price of the next step before you spend anything. For one paper it is about \$0.02:

```
research-companion cost-estimate
```

Step 3 - Build. The AI reads your paper once and extracts its concepts, claims, and references into the knowledge graph. The result is cached - you never pay for the same paper twice:

```
research-companion build --provider openai
```

Step 4 - Review, with the live screen. This runs all six helpers and opens a page in your browser where you watch them work in real time: the final report is saved as a file you can keep or share:

```
research-companion review local:<your-id> --serve --report my-review/
```

Step 5 - Answer your reviewers (when you have reviews):

```
research-companion rebuttal local:<your-id> --reviews reviews.txt
```

Try it with no account at all - the built-in demo uses bundled example data, needs no key and no internet, and finishes in about 30 seconds:

```
python examples/demo_offline.py
```

5. The screens you will see

The live dashboard (opens at <http://127.0.0.1:8501> when you use `--serve`). One card per helper showing its status - running, done, or FAILED - plus a feed of findings as they happen: the graph being built, a bad reference being caught, a novelty verdict arriving. It stays open after the run so you can look around; press Ctrl+C in the terminal to close it.

The report (the `my-review/report.html` file). A single page you can open anywhere, email, or archive. It shows every helper's results: your references with verdicts and reasons, a table of your claims with novelty verdicts, the confidence cards, and benchmark suggestions. Helpers that failed are shown in red with the reason - a partial report is still a useful report.

The graph view (run: `research-companion view`). An interactive map of everything the tool has read - papers, concepts, methods - where connected things sit together. Most useful once you have added several papers.

6. How to read the results

Citation verdicts:

verified	A live database record matches your reference. Good to go.
suspect	Something matches but something is off - the DOI disagrees, the authors do not match, or the title is close but not right. Check it by hand.
unverified	No plausible record exists in CrossRef or OpenAlex. Either the reference is malformed - or it was invented. Investigate before submitting.

Novelty verdicts: **novel** (nothing similar found), **incremental** (builds on known work), **overlaps** (a named prior paper does something very similar), **anticipated** (a prior paper already did this). Each verdict names the closest prior work, and shows whether the supporting quote from your paper was verified word-for-word (*quote-verified*) or not (*quote not found* - the AI paraphrased; trust it less).

Confidence scores like **0.72 +/- 0.15**: the first number is the score (closer to 1 is better); the second is the uncertainty band - how sure the tool is about its own score. A wide band (like +/- 0.50) means little evidence was available: treat the score as a hint, not a fact. The three ingredients of every score are printed with it.

Rebuttal honesty badges: **[OK all quotes verified]** - every quoted passage really appears in your paper. **[CHECK n unverified span(s)]** - some quoted text was NOT found in your paper; read before sending. **[NOTE reply cites no paper quotes]** - the reply is reasonable but quotes nothing checkable; verify its claims yourself.

7. What it costs and what stays private

- **Money:** reading one paper (the build step) costs roughly \$0.02 in AI usage; a full review adds a few cents more per claim. The citation check, the report, and the offline demo cost nothing. cost-estimate always tells you before you spend.
- **Privacy:** everything is stored on YOUR computer (in `~/research-companion`). Paper text is sent only to the AI provider you configured (OpenAI or Anthropic) when you run build, review, or rebuttal - never to the tool's authors. The offline demo sends nothing anywhere.
- **Audit trail:** every run writes a log file listing exactly what each helper did and found - your black-box recorder if you ever wonder what happened.

8. When things go wrong (they sometimes do)

- **A helper says FAILED with '429':** a free scholarly database (arXiv, Semantic Scholar) is rate-limiting - politely telling everyone to slow down. The other helpers still finish; just retry in a few minutes. The tool automatically switches to a backup database (OpenAlex) for prior-art search when this happens.
- **The AI key is rejected:** check three things - you saved the `.env` file, the key has no quote marks around it, and it is the full key (they are about 160 characters long).
- **All my citations show 'unverified':** does your PDF actually contain a bibliography? A draft exported without its reference list gives the citation detective nothing to check - it will honestly tell you exactly that.
- **Something odd happened:** open the newest file in `~/research-companion/runs/` - the audit log records every step, including the exact error and which helper hit it.

9. What it cannot do (honest limits)

- Quote checking is word-for-word: a made-up *paraphrase* of your paper will not be caught, only flagged when quoted. 'Verified' means the quote exists - not that the claim is true.
- Novelty verdicts are only as good as the related work found. If the literature search returns poor matches (it happens with generic titles), verdicts lean optimistic - always read the 'closest prior' list yourself.
- The databases it checks (CrossRef, OpenAlex, Semantic Scholar, arXiv) cover most of science well, but coverage is weaker for non-English venues and very new papers.
- It advises; it does not decide. Treat every verdict as a well-organized second opinion.

10. Get it

```
git clone https://github.com/Laraib-Hasan/Research-Companion
cd Research-Companion
pip install -e "[demo]"
python examples/demo_offline.py # 30-second tryout, no key needed
```

Free, open source (MIT license), runs on Windows, macOS, and Linux. Questions and issues: the GitHub page above.